

Read the Room: Grasp-Aware Planning and Learned Control for Multi-Robot Transport

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Abstract—Shared policy weights do not determine how robots coordinate through a common load. We study the planning and execution interfaces of two wheeled bimanual carriers using grasp-conditioned assembly geometry and shared learned local control. Assigned grasps and attained carrier placement define aperture constraints and object-level MPC references; a supervisor selects lift and transport phases. Including the carriers narrows a recorded beam’s admissible yaw from 25.8° to 5.5° . A frozen ten-scene simulation test with independent activation delays completes 35/40 missions; all failures precede verified-route execution. A later twelve-scene cohort completes 46/48 with a different actor and grasp configuration. Fixed-grasp spatial probes extend the reference to object roll within a 90° operating envelope: a geometric teacher reaches 44.29° and 88.29° for 45° and 90° targets, while the separate learned actor fails its 45° probe. These results separate geometric feasibility, supervised coordination and controller limitations. Training centrally pools teacher-labeled local observations; execution shares object references and readiness through phase/height commands. The evidence characterizes simulated transport under welded retention, without establishing a learning advantage or minimum communication.

I. INTRODUCTION

Can scaling a single-robot policy also produce a collaborating team? Larger datasets and shared models broaden the tasks and embodiments served by one policy [1]. Yet when copies run on different robots, will their actions complement one another, or must they communicate what they intend to do? Shared weights provide a common action rule; different observations and physical states still leave the team to coordinate intent, timing and support. Recent learned carrying and team policies make this distinction concrete [2]–[4].

Consider two robots carrying a couch through a narrow doorway. They must reach separated grasps, acquire contacts and lift together before turning through the opening. A couch that fits may leave insufficient room for its carriers. Their attained layout can differ from the nominal grasp arrangement, and their articulated arms allow that layout to change during motion. The challenge therefore spans acquisition and navigation: a feasible object path must also accommodate the team that executes it.

Self-driving offers a useful comparison. Decentralized driving policies choose actions for individual vehicles while accounting for other road users [5]. Their decisions interact, but their bodies are ordinarily mechanically separate. Robots holding a couch share a dynamical system: one robot’s acceleration changes the other’s load, and one robot’s lift moves the other’s grasp target. Local action selection must therefore remain compatible with both the intended motion and changing physical support.

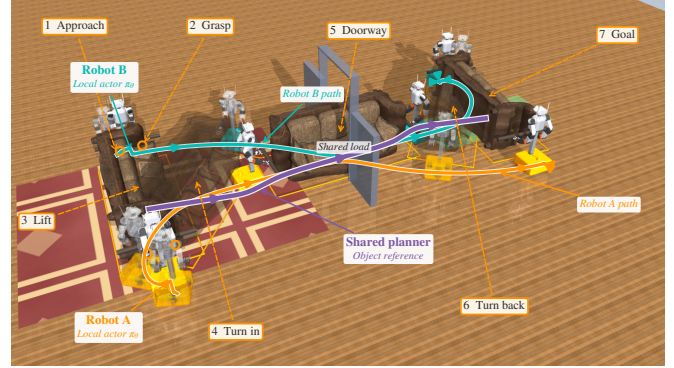


Fig. 1. Cooperative doorway transport with a shared planner and identical local policies.

Human collaboration suggests a possible interface. Partners can agree to “turn through the doorway” while adjusting their movements through vision and touch. Haptic feedback can improve coordination [6]; task-level agreement accompanies continuous physical feedback. For robots, this motivates sharing goals, grasp roles and motion constraints while each robot determines its own base, arm and lift actions. How much additional information is needed when a partner arrives late or changes support remains an open question.

We study *grasp-conditioned shared motion*: the task specifies where the object should go, while the attained carrier layout determines which motions are admissible. This differs from following an inferred partner wrench, as in PAINT [7]. The central interface maps assigned grasps and post-lift base placement to an assembly footprint, aperture-aware search and object-level MPC. Identical local actors track the resulting reference; a supervisor supplies phase and height. A separate spatial interface commands fixed-grasp roll within 90° . The contribution is the connection between acquired geometry, shared motion and execution conditions, rather than a new intent estimator or locomotion backbone.

Our contributions are:

- **Grasp-conditioned geometric feasibility:** an attained-formation aperture model and a displacement bound linking carrier drift to clearance.
- **A shared-reference control formulation:** common object motion and supervised progression drive identical local actors, with a separate fixed-grasp spatial reference and an explicit 90° roll envelope.
- **An evaluation across acquisition, transport and reorientation:** frozen 40- and 48-mission cohorts, recorded spatial probes and force-response diagnostics, with controller and contact-model differences retained.

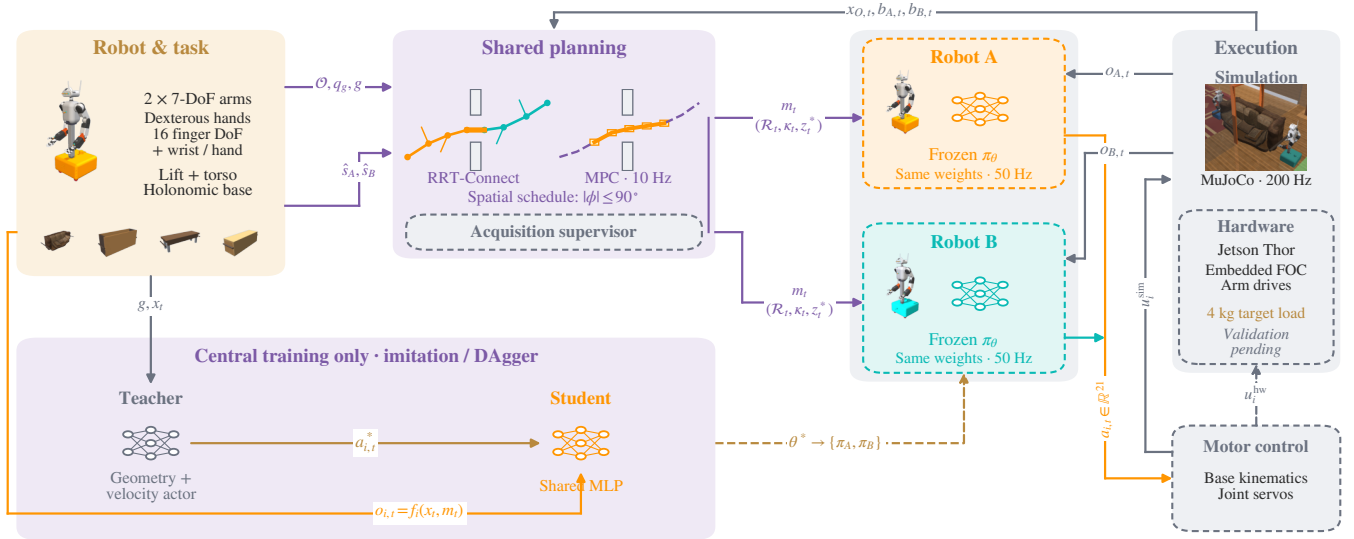


Fig. 2. Shared planning, centralized imitation and local execution. Signals follow Eq. (15); $\hat{s}_i$ are carrier offsets, u_i^{sim} servo targets and u_i^{hw} impedance targets. Dashed weights transfer offline; hardware validation is pending.

The experiments use idealized contact-gated welded retention and test new seeds within the training scene families. They do not establish superiority over a matched geometric controller, generalization to unseen layouts, or minimum communication. Communication amount, content and task-level agreement are discussed as subsequent research questions.

II. RELATED WORK

a) Physical interaction and inferred intent: Cooperative transport can be organized around following a partner’s guidance or tracking an agreed object motion. In the former setting, physical interaction carries information about the intended motion: role-based learning treats a partner’s actions as implicit communication [8], and haptic feedback contributes to human coordination [6]. The observation interface determines how that information reaches the controller. PAINT [7] represents intent as planar force and yaw torque. Its privileged teacher observes the interaction wrench; a student estimates it from arm proprioceptive history and learns with wrench regression and KL-regularized teacher guidance. A frozen locomotion policy executes the resulting base commands. This separates intent inference from locomotion and supports deployment without explicit payload tracking. Our actors instead receive object state and a shared geometric reference; we do not estimate a partner-applied wrench. The complementary question is how the attained grasp and carrier arrangement constrain the motion that a team can jointly track.

b) Shared policies and contact coordination: Broad individual skills provide a foundation for teamwork. TidyBot++ supports mobile-manipulation demonstration collection [9], while CrossFormer unifies heterogeneous observations, actions and control rates [1]. CHORUS adapts a shared VLA to concurrent collaboration using local visual observations and robot identity/role prompts [4]. These approaches make observation and timing part of the coordination problem even when weights are shared. Physical contact further constrains

the interface: decMBC conditions local control on the geometry of rigidly attached carriers [2]; decPLM maintains surface contact through a learned arm/base hierarchy and constellation rewards [3]. Its initial-only versus continuous contact-frame comparison links information requirements to payload shape and load. Sequential Asymmetric Imitation trains against progressively more realistic partner behavior, connecting interaction experience to learned waiting and yielding [10]. Our frozen evaluations retain an explicit acquisition supervisor; they do not establish the same local readiness or partner-inference capability.

c) Geometry and explicit shared references: Paired-grasp planning couples grasp selection to cooperative transport [11], while semi-coupled planning supplies a common object path with local redundancy resolution [12]. Distributed MPC exchanges predictions to coordinate coupled motion [13]; learned encodings compress such messages while retaining trajectory reconstruction [14]. A task goal, an object reference and a predicted partner trajectory therefore provide different information. Our formulation fixes grasp assignments, measures carrier offsets after lift, and uses their assembly footprint for aperture-constrained planning and object-level MPC. A displacement bound relates formation drift to reduced-model clearance, while separate spatial probes expose the limits of fixed-grasp roll. This focuses the contribution on the geometry and progression of a shared reference; a matched study is still needed to determine how much of that reference can be removed.

III. GRASP-AWARE PLANNING

The task specifies an object, assigned hand–rail contacts, a static map and a goal pose. Two wheeled bimanual robots acquire and lift the object before planning its transport. The attained carrier layout defines a planar assembly model for global search; object-level MPC supplies a common motion reference to the local actors (Fig. 2).

A. Grasp geometry as a team constraint

Let $q = (t, \theta) \in \text{SE}(2)$, $t = (x, y)$, denote object pose and $s_i = (u_i, v_i)$ the nominal position of base i in the object frame. With payload rectangle $\mathcal{P}$ of dimensions (L, D) and base-disc radii r_i ,

$$\mathcal{C}(s) = \mathcal{P} \cup \bigcup_{i=1}^2 B(s_i, r_i), \quad \mathcal{C}_q = t + R(\theta)\mathcal{C}(s). \quad (1)$$

Calibrated object profiles specify reachable grasps. Base offsets s_i are measured after lift and held fixed for route planning. This is a formation approximation: articulated arms allow the physical bases to move relative to the object. For obstacles $\mathcal{O}$ and signed clearance d , the configuration-space problem [15] is

$$\begin{aligned} \gamma : [0, 1] &\rightarrow \text{SE}(2), \quad \gamma(0) = q_{\text{lift}}, \quad \gamma(1) = q_g, \\ d(\mathcal{C}_{\gamma(\tau)}, \mathcal{O}) &\geq \varepsilon \quad \text{for all } \tau \in [0, 1]. \end{aligned} \quad (2)$$

The collision model excludes supports below a specified carrying-height bound and represents the payload and bases, not the articulated swept volume.

The sensitivity to formation drift can be stated without assuming rigid attachment of the bases. Let s^0 denote planning offsets and $\delta_s(t) = \max_i \|s_i(t) - s_i^0\|$. At the same object pose q , translation of each base disc by at most δ_s gives

$$\text{dist}(\mathcal{C}_q(s(t)), \mathcal{O}) \geq \text{dist}(\mathcal{C}_q(s^0), \mathcal{O}) - \delta_s(t). \quad (3)$$

The triangle inequality gives this reduced-footprint bound. Clearance above δ_s suffices at the same object pose; articulated swept volume and path error are excluded. The test is retrospective, not enforced online.

For a centered opening of width W , define $w_{\mathcal{P}} = L|\sin \theta| + D|\cos \theta|$ and

$$\begin{aligned} w_i(\theta; s) &= 2(|u_i \sin \theta + v_i \cos \theta| + r_i), \\ \bar{w}(\theta; s) &= \max\{w_{\mathcal{P}}, w_1(\theta; s), w_2(\theta; s)\}. \end{aligned} \quad (4)$$

The aperture condition $\bar{w} \leq W - 2\varepsilon$ is exact for mirrored, equal-radius layouts and conservative otherwise. Yaw scanning and boundary bisection identify admissible intervals.

The corresponding gate-pose set couples orientation and lateral placement:

$$\mathcal{G} = \left\{ (x_d, y, \theta) : |y - y_d| \leq \frac{W - \bar{w}(\theta; s)}{2} - \varepsilon \right\}. \quad (5)$$

Here (x_d, y_d) is the opening center; a negative bound excludes that yaw. This sampling constraint does not certify clearance elsewhere. For the recorded beam handoff, the object-only admissible yaw window is 25.8° ; including the carriers reduces it to 5.5° (supplementary geometric illustration).

Gate-biased RRT-Connect [16] samples the opposite root, aperture set and workspace with probabilities $(0.10, 0.35, 0.55)$. Steering uses $d_q = \|t_a - t_b\|_2 + 0.6|\text{wrap}(\theta_a - \theta_b)|$ and 0.35 m-equivalent extensions. Search permits three seeded attempts of 20,000 iterations, followed by shortcutting and clearance-checked smoothing. Swept collision checks use 5 mm increments and a 22 mm margin; an independent 10 mm-resampling verifier requires $\varepsilon = 20$ mm along the delivered path, including joins. These finite-resolution checks implement Eq. (2); failed verification rejects

the route.

B. Shared receding-horizon object reference

Object MPC [17] predicts q_k under object-frame twist $\nu_k = (v_x, v_y, \omega)$ with timestep h :

$$q_{k+1} = f(q_k, \nu_k) = q_k + h \text{diag}(R(\theta_k), 1)\nu_k. \quad (6)$$

At each solve, measured object pose and base offsets update the model. Offsets are parameters, not control decisions. Monotone measured path progress selects $\bar{q}_{1:N}$, so elapsed time alone cannot advance the reference during a stall. With $e_k = q_k - \bar{q}_k$ and locally unwrapped yaw, successive convexification solves

$$\begin{aligned} \min_{q_{1:N}, \nu_{0:N-1}, \sigma \geq 0} \quad & \sum_{k=1}^{N-1} \|e_k\|_Q^2 + \|e_N\|_{Q_f}^2 \\ & + \sum_{k=0}^{N-1} \|\nu_k\|_{R_\nu}^2 + \lambda \|\sigma\|_1 \\ \text{s.t.} \quad & q_0 = \hat{q}, \quad q_{k+1} = A_k q_k + B_k \nu_k + c_k, \\ & a_{jk}^\top q_k + \sigma_{jk} \geq b_{jk} + \varepsilon, \\ & |\nu_k| \leq \nu_{\max}, \quad |\nu_k - \nu_{k-1}| \leq h \dot{\nu}_{\max}, \\ & \|\nu_{xy,k} + \omega_k J \hat{s}_i\|_\infty \leq 1.2 \text{ m/s}. \end{aligned} \quad (7)$$

Here J rotates by 90° and $\hat{s}_i$ is the measured base offset. Affine dynamics approximate Eq. (6); separating planes approximate obstacle clearance with individual nonnegative slacks σ_{jk} . The frozen profile uses $N = 20$, $h = 0.1$ s, $Q = \text{diag}(12, 12, 30)$, $Q_f = 40Q$, $R_\nu = \text{diag}(0.6, 0.6, 0.8)$ and $\lambda = 4000$. Two warm-started convexification passes use OSQP [18]. Limits are 0.12 m/s per translation axis and 0.10 rad/s yaw, with slew bounds of 0.15 m/s² and 0.10 rad/s². The first control is accepted only if the solve succeeds, the command is finite, and maximum slack is at most 10 mm; otherwise the bridge commands zero twist. Execution clearance is measured independently of the QP slacks.

For world-frame station offset ρ_i from the object center, the common reference induces rigid-motion feedforward

$$v_i^{\text{ff}} = R(\theta)(v_x, v_y)^\top + \omega \hat{z} \times \rho_i, \quad \omega_i^{\text{ff}} = \omega. \quad (8)$$

Each robot transforms this reference into its base frame and adds bounded learned residuals. Both planners use the same assembly geometry and requested clearance. Reference updates preserve the continuous physical state.

C. Fixed-grasp object reorientation

Spatial pose tracking extends the reference interface; it does not replace planar RRT/MPC with a collision-free SE(3) search. Let $T_O = (R_O, p_O) \in \text{SE}(3)$ and let $\bar{T}_{ih}$ be the fixed object-to-hand transform for robot i , hand $h \in \{L, R\}$. The reference satisfies

$$\begin{aligned} T_{ih}^*(t) &= T_O^*(t) \bar{T}_{ih}, \\ R_O^*(t) &= R_{O,0} \exp(\phi(t)[e_x]_\times). \end{aligned} \quad (9)$$

where $R_{O,0}$ is the lifted orientation and e_x the object's long axis. Upper/lower handles provide distinct hand targets during roll. We adopt the fixed-grasp physical operating envelope

$$|\phi(t)| \leq \phi_{\max} = \pi/2, \quad \bar{T}_{ih}(t) = \bar{T}_{ih}(0). \quad (10)$$

Beyond 90° , continuing the same hand assignment twists the arms and requires switching hands along the handles. Such

contact reassignment and regrasp planning are outside this paper. This mechanism-specific envelope is not a guarantee of reachability for every pose within it. It bounds object roll relative to the lifted frame, not world-frame yaw during transport. Together, the planar and spatial tests examine cooperative translation, yaw and out-of-plane rotation (Fig. 3); independent pitch-axis control is not evaluated.

The high-level reference specifies a clearance lift followed by roll. Over a segment beginning at t_a with duration τ , set $u = (t - t_a)/\tau \in [0, 1]$ and interpolate

$$\begin{aligned} s(u) &= 10u^3 - 15u^4 + 6u^5, \\ p^*(u) &= p_a + s(u)(p_b - p_a), \\ R^*(u) &= \exp(s(u) \log(R_b R_a^\top)) R_a. \end{aligned} \quad (11)$$

Differentiation supplies the centroid twist. Unlike planar MPC's measured path progress, these pose probes use a timed schedule after supervised lift. The local actor receives pose/twist errors and hand targets in its base frame; the geometric teacher instead maps hand errors to bounded Jacobian rates with joint-limit and selected self-clearance constraints. Neither controller chooses the requested endpoint. Large commanded roll replaces the planar upright-object guard with pose-tracking and contact guards; the 2° base-inclination limit remains.

IV. LOCAL CONTROL AND COORDINATION

A. Robot platform and motor interface

Each carrier has a holonomic three-module steering/drive base, a two-stage lift, a pitching torso, two 7-DoF arms and two hands with 16 finger DoFs and an additional wrist each (Fig. 2A). The planar policy commands three base and eighteen upper-body coordinates; hand articulation uses stage-dependent closure rather than learned finger-level actions. The hardware interface uses Jetson Thor for local inference and CAN arm-drive commands $(q^d, \dot{q}^d, K_p, K_d, \tau^{\text{ff}})$:

$$\tau^d = \text{clip}(K_p(q^d - q) + K_d(\dot{q}^d - \dot{q}) + \tau^{\text{ff}}, -\tau_{\text{max}}, \tau_{\text{max}}). \quad (12)$$

Embedded field-oriented control (FOC) regulates arm-motor current; MuJoCo uses its actuator model instead. The 4 kg hardware target load is distinct from the evaluated 1–2 kg simulation loads. Hardware system performance and that load target are not validated by the reported experiments.

B. Shared policy and local observations

In the frozen planar cohort, each robot executes $a_i = \pi_\theta(o_i)$ at 50 Hz with identical weights; MuJoCo dynamics run at 200 Hz [19]. The 157-dimensional observation includes joint state and tracking errors, base motion and gravity, object-relative pose/twist, grasp geometry, palm-target and docking errors, phase/height references, previous action and commanded motion. It excludes partner joint/velocity streams. Simulation supplies object state and six clipped constraint-multiplier channels for grasp feedback.

For base position p_i and yaw ψ_i , relative features are

$$\begin{aligned} \tilde{p}_i &= R(-\psi_i)(t - p_i), & \tilde{s}_i &= R(\theta - \psi_i)s_i, \\ c_i &= (\cos 2(\theta - \psi_i), \sin 2(\theta - \psi_i)). \end{aligned} \quad (13)$$

The double-angle feature represents footprint symmetry; assigned hand targets retain contact identity. This common coordinate convention permits weight sharing between opposed carriers without assuming policy equivariance. The 512–256–128 ELU MLP produces three base and 18 upper-body actions. Arm targets are posture residuals; lift actions specify a bounded 0.06 m/s rate during lifting and position residuals around the attained carrying posture during transport.

C. Acquisition and asynchronous progression

Assigned rails define the approach targets. The geometric teacher stages, docks and refines wrist pose within 40 mm. Panel/beam profiles use 0.18 m hand spacing instead of 0.40 m and reserve 60 mm downward lift travel. Robot i attaches when both assigned palms are within 35 mm and 30° of their targets, base speed is below 0.10 m/s, and both palm/finger contacts apply more than 0.1 N compression to the correct rails. Weld constraints then retain the attained transforms [20]; subsequent frictional slip is not modeled.

One actor starts immediately and the other after $\Delta \in [0, 8]$ s; the delayed robot holds its initial command until activation. Let b_i indicate attachment and z_0 initial object height. The attached robot's height reference is

$$z_i^* = \begin{cases} z_0 + 0.04 \text{ m}, & b_i = 1, \ b_1 b_2 = 0, \\ z_0 + 0.12 \text{ m}, & b_1 b_2 = 1. \end{cases} \quad (14)$$

During early lift, the base holds and upward action is clipped at a 40 mm centroid rise. Remaining support permits this partial lift while the late robot tracks its moving grasp targets. Joint attachment releases the full lift. Transport begins after height error remains below 20 mm and object roll/pitch below 0.06 rad for 25 control steps (0.5 s). The same actor then tracks MPC references without resetting the physical state. Gated-start comparisons instead synchronize controller availability and attachment.

D. Information and physical assumptions

Let x_t be the simulator state, $b_{i,t}$ attachment status, and $m_t = (\mathcal{R}_t, \kappa_t, z_t^*)$ the shared reference: motion, phase and height. The evaluated information flow is

$$\begin{aligned} m_t &= \mathcal{H}(x_{O,t}, b_{1,t}, b_{2,t}, g), \\ o_{i,t} &= f_i(x_t, m_t), & a_{i,t} &= \pi_\theta(o_{i,t}), \\ \mathcal{D} &= \bigcup_{i,t} \{(o_{i,t}, a_{i,t}^*)\}, \\ \theta^* &= \arg \min_{\theta} \sum_{(o, a^*) \in \mathcal{D}} \|\pi_\theta(o) - a^*\|_W^2. \end{aligned} \quad (15)$$

Here g is task geometry, $\mathcal{R}_t$ the motion reference and W a diagonal action-weight matrix. Central training pools teacher-labeled samples from both robots; teacher labels may use simulator geometry and supervised progression. Student execution retains identical weights and normalization, but no teacher labels, optimizer state or partner joint/velocity streams.

Shared memory delivers simulator object state, 10 Hz planar MPC references and supervisor-selected phase/height to the 50 Hz actors. Spatial probes sample the common

TABLE I

PER-ROBOT ACQUISITION REWARDS IN THE SEPARATE PPO COMPARISON. RATES ARE MULTIPLIED BY $\Delta t = 0.02$ s; EVENT REWARDS ARE ADDED ONCE. THE FROZEN FULL-SYSTEM ACTOR INSTEAD MINIMIZES IMITATION LOSS.

Term	Rate / event value
Approach progress	$6 \text{ clip}((d_{t-1} - d_t)/\Delta t, -0.1, 0.1)$
Proximity	$2 \exp[-(d_t/0.10)^2]$
Alignment, original	$\exp[-(\alpha_t/0.35)^2]$
Alignment, gated	$\exp[-(\alpha_t/0.35)^2 - (d_t/0.15)^2]$
Lift height	$10H(z_t - z^*)$
Lift levelness	$3 \exp[-\ \eta_t\ ^2/0.15^2]$
Approach / lift costs	$-0.5 - 4(\beta_i/\beta_{\max})^2$
Transition / failure	+2 each at both-attached and lift-complete; -5 on physical termination.

pose schedule at the control rate. These references convey joint readiness even without direct actor-to-actor messages. Configured update rates are not measured network traffic. The separate PPO and local-history studies use centralized critics during training only; their information and results are specified in the supplement.

The frozen planar actor's six clipped feedback channels aggregate equality constraint multipliers, not calibrated wrist wrenches. Welds can sustain tensile forces and moments without frictional retention limits. The supplement defines the feature scaling and physical assumptions; no load-capacity or actuator-saturation claim is made.

V. LEARNING OBJECTIVES AND TRAINING PROTOCOLS

The frozen full-mission policy is fitted by imitation, Eq. (15). Reward optimization belongs to the separate pickup PPO and native-contact development branches; their objectives and results must not be interchanged.

A. Archived pickup PPO

This experiment is not an update to the frozen full-mission actor. It compares pickup reward variants, not end-to-end transport performance. Two separate PPO branches each run 65,536 steps with an asymmetric 181-input critic [21], [22]. They use generalized advantages [23], a 0.3 KL anchor, value coefficient 0.5 and zero entropy coefficient. Table I defines their acquisition rewards; the variants differ only in distance gating of orientation reward. Transport rewards retain common command-tracking, posture, grasp and regularization terms, detailed in Table II. All evaluations use deterministic actor means.

For maximum hand-position/orientation errors d_t, α_t , object roll/pitch η_t and base inclination β_i , the normalized height term is

$$\Phi(e) = (e^{-(e/0.08)^2} + 0.5e^{-(e/0.25)^2})/1.5, \quad (16)$$

$$H(e) = \max\left\{0, \frac{\Phi(e) - \Phi(e_0)}{1 - \Phi(e_0)}\right\},$$

where $e = z - z^*$ and $e_0 = z_{\text{rest}} - z^*$. At zero initial error, $H = \Phi$. The inclination scale is $\beta_{\max} = 2^\circ$.

Both branches use one environment, horizon 128 and 512 updates; the learning rate is 3×10^{-5} , with three epochs, four

minibatches, clip ratio 0.2, $\gamma = 0.99$ and GAE $\lambda = 0.95$. Normalization is frozen and the critic receives 32 warmup updates. The network has ELU hidden layers 512–256–128. Mass is sampled from $U[0.75, 3]$ kg, dimensions from $U[0.9, 1.1]$, base offsets from 0.08–0.65 m, headings within $\pm 30^\circ$, and activation delays within 0–2 s. COM perturbations are $\pm(0.05, 0.12, 0.05)$ m and servo-gain factors 0.85–1.15. Automatic curricula are disabled.

Frozen pickup outcomes are 40/40 for the original reward and 39/40 for the distance-gated reward. Both stochastic training branches finish at 21/37 completed episodes. There is no demonstrated improvement from reward gating, and neither result establishes full-mission reliability.

B. Complete archived transport objective

For $K_\sigma(e) = \exp(-\|e\|^2/\sigma^2)$, the command-normalized kernel is

$$b = K_\sigma(c), \quad k = K_\sigma(x - c),$$

$$T_\sigma(x, c) = \frac{k - b}{\max(1 - b, 0.1)} + \left[1 - \min\left(\frac{1 - b}{0.1}, 1\right)\right] k. \quad (17)$$

Table II specifies every transport term in the archived PPO environment. Approach/lift replace this sum with Table I; they do not add both objectives.

C. Native-contact development objective

The native-contact branch removes grasp welds and artificial attachment springs. Its per-robot objective combines progress, support and regularization with measured finger-contact costs:

$$r_{i,t} = \sum_{k \in \mathcal{K}_{\text{task}}} w_k f_{k,i,t} + \sum_{k \in \mathcal{K}_{\text{contact}}} w_k f_{k,i,t}. \quad (18)$$

Tables III and IV report guided-continuation (G) and the newer hand-target configuration (H) side by side; neither is retroactively assigned to earlier trials. Rates include $\Delta t = 0.02$ s; progress, event, work and effort increments do not receive a second timestep factor. Both actors receive local observations; training aggregates their trajectories centrally. This objective does not establish native-contact reliability.

For hands $h \in \{L, R\}$, $\langle x_h \rangle = \frac{1}{2} \sum_h x_h$, d_h is palm-to-target distance, c_h closure, q_h raw contact qualification, a_h retained contact status, f_h firmness and m_h friction margin. The acquisition potential and support gate are

$$\Phi_i = \sum_h \left[e^{-(d_h/0.04)^2} \text{clip}(c_h/c^*, 0, 1) + a_h \right],$$

$$G(v) = \begin{cases} v, & \text{supported at both interval endpoints,} \\ \min(v, 0), & \text{otherwise.} \end{cases} \quad (19)$$

The position-rotation error uses $D = \text{diag}(1, 1, 1, \rho, \rho, \rho)$, $\rho = 0.35$ m. Native contact shaping uses $\sigma_d = 0.06$ m, $\sigma_v = 0.05$ m/s and $F_{\max} = 40$ N; the separate load normalization uses $F_{\text{cap}} = 60$ N. Finger work is $\Delta W_i = \int \sum_j |\tau_j \dot{q}_j| dt$ and effort is $\Delta E_i = \int \sum_j \tau_j^2 dt$. The contact-frame cost uses four landmarks $\{0, 0.04e_x, 0.04e_y, 0.04e_z\}$ per palm relative to the assigned or captured object frame. Recorded contact

TABLE II

ARCHIVED PPO TRANSPORT OBJECTIVE. EVERY ROW IS MULTIPLIED BY $w_k \Delta t$; TERMINAL SEQUENCE FAILURE ADDS -5 ONCE. THESE REWARDS DO NOT OPTIMIZE THE FROZEN IMITATION ACTOR.

Term	Weight	Raw contribution f_k	Definition / gate
Command task	10	$e^{-\ (\xi - \xi^*) / (0.15, 0.15, 0.2)\ ^2}$	Defines P ; planar twist.
Linear tracking	3	$T_{0.25}(v, v^*)$	No gate.
Angular tracking	2	$T_{0.35}(\omega, \omega^*)$	No gate.
Height	2	$H(z - z^*)$	Multiplied by P .
Level	1	$K_{0.15}(\eta)$	Multiplied by P .
Grasp	1	$\mathbf{1}[\text{engaged}_i]$	Multiplied by P .
Base inclination	4	$-(\max_j \beta_j / (\pi/90))^2$	No gate.
Torso posture	0.5	$K_{0.261799}(\psi_i)$	Multiplied by P .
Arm symmetry	0.5	$K_{0.15}(e_{\text{sym},i})$	Multiplied by P ; metric hand-symmetry error.
Contact constellation	1	$-E_{c,i} / 0.05^2$	Mean squared palm-landmark error.
Base constellation	1	$-\mathbf{1}[\text{engaged}_i] E_{b,i} / 0.05^2$	Mean squared base-landmark error.
Residual magnitude	2	$-\ a_{0:3}\ ^2 - 0.1\ a_{3:}\ ^2$	Base and upper-body action costs.
Object tilt	0.5	$-\min(\ \eta\ _\infty / 0.15^2, 4)$	No gate.
Action change	0.005	$-\ a_t - a_{t-1}\ ^2$	No gate.
Drop	50	$-\mathbf{1}[z < z_{\text{rest}} - 0.12]$	No gate.

TABLE III

COMPLETE NATIVE-CONTACT TASK OBJECTIVE. WEIGHTS G/H DENOTE GUIDED CONTINUATION / THE NEWER HAND-TARGET CONFIGURATION. ZERO DISABLES A TERM; TIMESTEP AND EVENT CONVENTIONS ARE EXPLICIT.

Term	G	H	Raw contribution f_k	Definition
Grasp progress	20	20	$\text{clip}(g_{t-1} - g_t, -.05, .05)$	Metric grasp-error reduction.
Rotation progress	0	6	$\text{clip}(\alpha_{t-1} - \alpha_t, -.02, .02)$	Angular grasp-error reduction.
Contact potential	5	5	$\Phi_{i,t} - \Phi_{i,t-1}$	Distance, closure and retained contact.
Grasp loss	6	6	$-N_{\text{release}}$	Excludes injected self-opening.
Proximity	1	1	$\Delta t e^{-(g/.08)^2}$	Dense approach reward.
Supported progress	100	100	$G(\Delta s)$	Path progress.
Goal progress	150	200	$G(\ De_{t-1}\ - \ De_t\)$	Translation and rotation.
Supported lift progress	0	200	$G(e_{z,t-1} - e_{z,t})$	Supported height-error reduction.
Stable goal	2	2	$\Delta t \mathbf{1}[\text{good}]$	Supported stable-goal condition.
Unsupported travel	20	20	$-\Delta \ell_{xy} \mathbf{1}[\text{unsupported}]$	Horizontal object displacement.
Anchor slip	2	2	$-\Delta t \min(\delta, .1) \mathbf{1}[\text{engaged}_i]$	Captured-frame grasp drift.
Load	0.2	0.2	$-\Delta t (\ F_i\ / \max(1, 2F_{\text{cap}}))^2$	Configured load normalization.
Object orientation	4	4	$-\Delta t \ e_R\ ^2$	Nearby-reference rotation error.
Torso posture	0.2	0.2	$-\Delta t \psi_i^2$	Local torso pitch.
Action change	0.01	0.01	$-\Delta t \ a_t - a_{t-1}\ ^2$	Normalized actor outputs.
Time	2	2	$-\Delta t$	Per-second cost.
Success	300	300	$\mathbf{1}[\text{success}]$	Terminal bonus.
Failure	500	500	$-\mathbf{1}[\text{terminated}]$	Physical termination penalty.

proxies are simulator diagnostics, not hardware-calibrated force-closure certificates.

For contact normals n_a on distinct finger bodies, the opposition score, normal-force total and force-weighted slip are

$$\begin{aligned}
 o_h &= \max_{a,b} \frac{1 - n_a^\top n_b}{2}, \quad F_{n,h} = \sum_a F_{n,a}, \\
 v_{\text{slip},h} &= \frac{\sum_a F_{n,a} v_{\text{slip},a}}{\max(F_{n,h}, 10^{-12})}, \\
 f_h &= \min(F_{n,h}/12, 1) o_h, \\
 m_h &= \text{clip}(1 - \max_a \chi_a, 0, 1), \\
 q_h &= \mathbf{1}[o_h \geq 0.7, F_{n,h} \geq 2, v_{\text{slip},h} \leq 0.04],
 \end{aligned} \tag{20}$$

where χ_a is contact friction utilization. With no contact, all quality quantities are zero. First acquisition requires 0.08 s qualification; retained status tolerates at most 0.02 s unqualified contact, unless normal force falls below 0.2 N. These temporal filters affect observation and reward bookkeeping,

not attachment forces.

The hand-target extension adds pose alignment, measured closure progress and relative held-hand motion, together with supported lift progress:

$$\begin{aligned}
 A_h &= e^{-(d_h/.03)^2 - (\alpha_h/.25)^2}, \\
 \Psi_i &= \langle A_h \text{clip}(\bar{c}_h/.6, 0, 1) \rangle.
 \end{aligned} \tag{21}$$

Here $\bar{c}_h$ is mean normalized MCP/PIP flexion of the non-thumb fingers, measured from joint encoders. Relative velocities subtract the object's rigid motion at the actual hand position. The configuration change also raises rotation/goal progress weights and disables the secure-contact cost; Tables III–IV retain every weight in both versions. No frozen outcome is attributed to these new terms without a matching configuration and checkpoint.

$$J_h = \text{clip}\left(\frac{\|v_{\text{rel},h}\|^2}{.03^2} + \frac{\|\omega_{\text{rel},h}\|^2}{.2^2}, 0, 1\right). \tag{22}$$

TABLE IV

COMPLETE NATIVE-CONTACT CONTACT OBJECTIVE. WEIGHTS G/H DENOTE GUIDED CONTINUATION / THE NEWER HAND-TARGET CONFIGURATION. ZERO DISABLES A TERM; TIMESTEP AND EVENT CONVENTIONS ARE EXPLICIT.

Term	G	H	Raw contribution f_k	Definition
First acquisition	8	8	ΔN_i	One event per hand's first qualified acquisition.
Distance	6	6	$-\Delta t \langle \tanh(d_h / \sigma_d) \rangle$	Metric palm-target separation.
Separation speed	3	3	$-\Delta t \langle \text{clip}(v_{\text{out},h} / \sigma_v, 0, 1) \rangle$	Outward relative velocity.
Secure contact	4	0	$-\Delta t \langle 1 - q_h \rangle$	Qualified opposing contacts.
Firmness	1.5	1.5	$-\Delta t \langle 1 - f_h \rangle$	Contact-quality proxy.
Contact slip	12	12	$-\Delta t \langle v_{\text{slip},h} \rangle$	Tangential slip speed.
Friction reserve	0.5	0.5	$-\Delta t \langle 1 - f_h m_h \rangle$	Firmness-weighted friction margin.
Excess force	2	2	$-\Delta t \langle \max(F_{n,h} / F_{\max} - 1, 0)^2 \rangle$	Normal-force overage.
Finger work	0.02	0.02	$-\Delta W_i$	Absolute actuator work increment.
Holding effort	0.002	0.002	$-\Delta E_i$	Squared-torque integral.
Premature closure	0.5	0.5	$-\Delta t \sum_h \mathbf{1}[d_{\text{nom},h} > .06] c_h^2$	Closure away from the nominal target.
Contact frame	100	100	$-\Delta t \sum_h \text{mean}_k \ p_h + R_h u_k - p_h^* - R_h^* u_k\ ^2$	Position and orientation via four landmarks.
Pose alignment	0	3	$-\Delta t \langle 1 - A_h \rangle$	Assigned or captured palm frame.
Aligned closure	0	12	$\Psi_t - \Psi_{t-1}$	Measured finger flexion, weighted by alignment.
Held-hand adjustment	0	1	$-\Delta t \langle a_h J_h \rangle$	Motion relative to the rigid object at the hand.

VI. EXPERIMENTAL EVALUATION

A. Evaluation scope

The original frozen cohort uses four held-out seeds per scene within the ten collection scene families and one trained actor. Later planar, spatial and local-history evaluations retain separate checkpoints and denominators. Neither unseen-family generalization nor training-seed robustness is tested.

B. Training protocol

The planar actor uses imitation and DAGger [24]: a geometric teacher labels acquisition/lift and a saved velocity actor labels transport. Three rounds collect 30 episodes each, first with the teacher and then with the student under teacher labeling. The 524,490-sample dataset includes 345,484 imported development samples; 86/90 collection episodes complete stable holding. Training centrally minimizes phase-balanced weighted action MSE across both robots. The actor receives no subsequent PPO update.

Collection randomizes mass $U[1, 2]$ kg, dimensions $U[0.95, 1.05]$, rail friction $U[0.9, 1.8]$, base offsets 0.08–0.50 m, headings $\pm 20^\circ$ and activation delays 0–8 s. Automatic curricula are disabled. Episodes truncate at 35 s or stop upon physical failure or completed collection. Furnished-scene collection holds after lift; RRT/MPC supplies navigation at test time. The supplement specifies imported-data provenance, feature rebasing, normalization, loss weights and fitting settings.

C. Mission test and outcome criteria

The frozen test comprises four prespecified seeds in each of ten scenes (Table V): nine doorway environments and an open arena. Each mission starts detached and has a 180 s simulation budget. Mass, dimensions, base offsets, headings and activation delays use the DAGger ranges above; object XY and yaw additionally vary by ± 0.10 m and ± 0.20 rad. Thus visually large payloads represent 1–2 kg loads. The paired MPC tests use a separate actor and domain (Table VI); their results are not pooled.

TABLE V

SCENE GEOMETRY AND FROZEN OUTCOMES: FOUR TRIALS PER SCENE.

Scene	Object	$L \times D \times H$ (m)	S (%)	T_L	T_G
Living	Sofa	$2.73 \times 1.12 \times 0.93$	75	9.0	98.3
Bedroom	Wardrobe	$2.10 \times 0.62 \times 0.95$	100	11.7	90.9
Dining	Table	$2.40 \times 0.76 \times 0.75$	100	8.6	94.9
Dock	Case	$2.80 \times 0.82 \times 0.80$	100	9.3	106.3
Warehouse	Crate	$2.20 \times 0.70 \times 0.65$	100	9.7	96.4
Gallery	Panel	$1.90 \times 0.09 \times 1.25$	50	11.6	93.1
Workshop	Beam	$3.20 \times 0.20 \times 0.20$	75	15.0	122.1
Office	Table	$2.40 \times 1.10 \times 0.74$	75	9.1	93.5
Apartment	Sofa	$2.10 \times 0.86 \times 0.78$	100	10.7	91.8
Arena	Sofa	$2.10 \times 0.90 \times 0.83$	100	11.3	40.3
All (40)		–	87.5	10.6	95.4

Nominal dimensions; variation $\pm 5\%$. T_L/T_G : median seconds to completed lift/first goal among passes. Simulation time; planning compute is excluded.

TABLE VI

EVALUATION PROTOCOLS AND ORIGINAL DENOMINATORS.

Test	Domain / comparison	Pass
Frozen missions	10 scenes; 1–2 kg; independent starts, $\Delta \leq 8$ s; 180 s.	35/40
Original MPC	8 scenes; 1.5 kg; gated starts, $\Delta \leq 0.8$ s; 180 s.	36/40
Aligned MPC	Same actor and 40 seeds as original MPC; geometry and margin change.	40/40

Let F denote physical termination at any point in a mission. Failure guards reject object heights outside $[z_{\text{rest}} - 0.12, z_{\text{rest}} + 0.30]$ m, object roll/pitch exceeding 0.6 rad, base inclination above 2° , or forbidden world contact. Base inclination is checked every physics substep. With final object position t_T , yaw error $\delta\theta_T$, completed-lift flag ℓ , attachment flags b_i , and minimum measured assembly clearance $d_{\min}$, mission success is

$$S = \mathbf{1}\{b_1 b_2 = 1, \ell = 1, \|t_T - t_g\| < 0.10, |\delta\theta_T| < 0.10, d_{\min} \geq 0, \neg F\}. \quad (23)$$

Position and angle tolerances are in meters and radians. Incomplete acquisition/lift and rejected routes count as failures. Planned clearance (20 mm) and measured execution clearance (0 mm) are separate criteria. All original trials remain in the denominator; Fig. 3 replays recorded states.

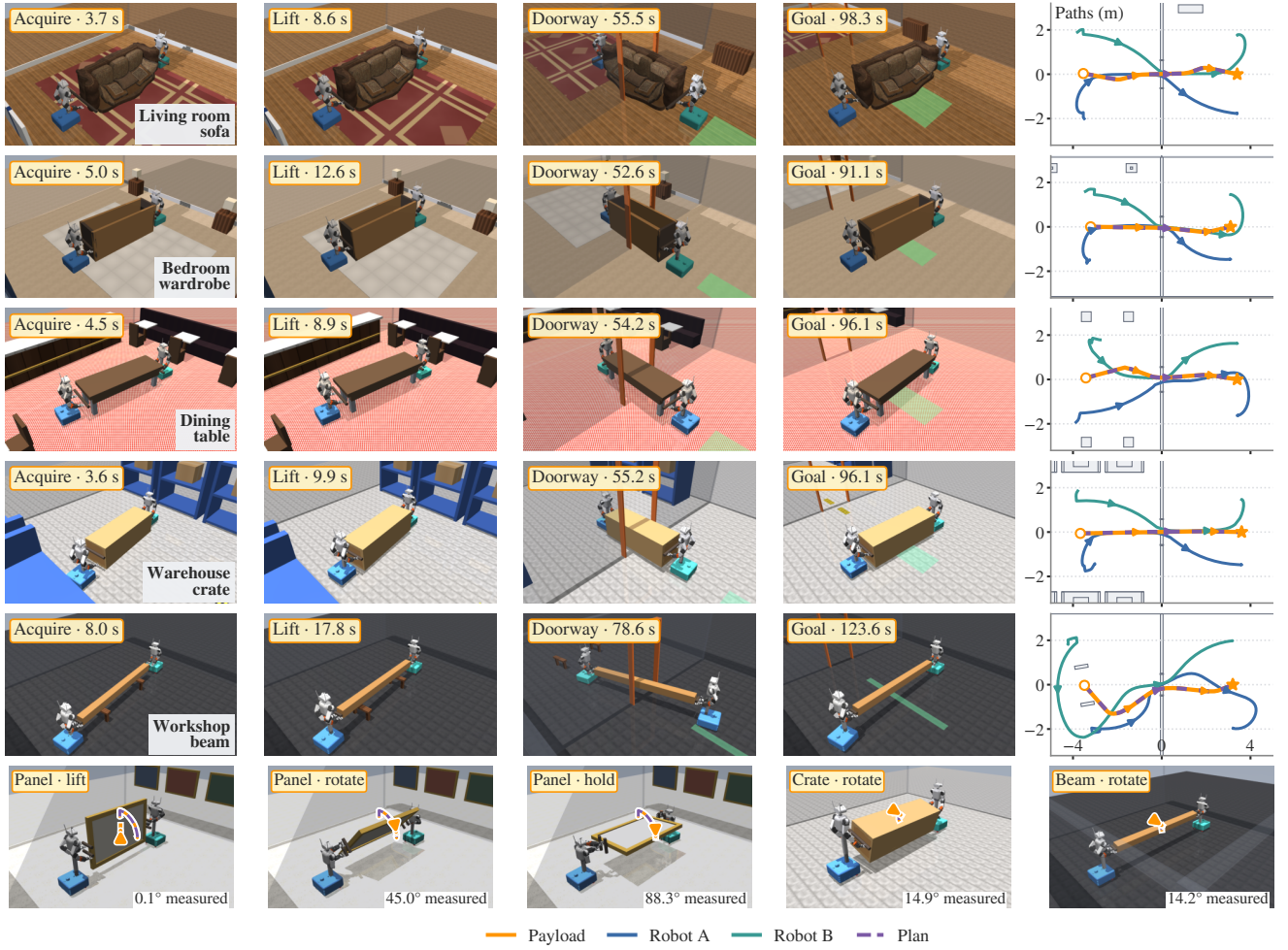


Fig. 3. Cooperative object-pose execution. Top five rows retain the original sofa, wardrobe, table, crate and beam missions: acquisition, lift, doorway passage, arrival and planar trajectories. Both position and yaw vary. Bottom row adds teacher-controlled out-of-plane reorientation: panel lift, intermediate/terminal roll, and single-bar crate/beam roll. Arrows show motion direction; purple dashed/orange solid paths are commanded/measured. Reorientation paths track a representative object-fixed point and labels give measured roll. Robot bodies retain their original materials; blue/teal bases identify Robot A/B. These are separate trials, not a combined tilt-through-door experiment. Table V retains all frozen mission outcomes.

Computation time is reported separately from simulation duration.

D. Geometric compatibility and asynchronous acquisition

The paired development test changes the local disc cover and 30 mm margin to the global rectangle model and 20 mm margin. Completion rises from 36/40 to 40/40, recovering four living-room stalls with the same actor (supplementary planning trace). Geometry, margin and archived implementation settings change together; this comparison diagnoses incompatibility rather than isolating a planning improvement.

With a 7.68 s activation gap, attachment occurs at 1.88/10.76 s and transport starts at 13.44 s (supplementary trace). Early lift accommodates a late partner through supervised progression; the delayed beam failure shows that acquisition geometry can still prevent completion.

A same-start panel replay (seed 142020) succeeds with the geometric teacher where the frozen actor tips the panel. This diagnostic motivates a matched full-cohort baseline; it does

not estimate overall teacher performance. The separate pickup PPO study and its reward equations are supplementary.

E. Reliability across acquisition and transport

The original frozen test completes **35/40 missions (87.5%)**. Failures comprise two panel tip-overs before grasp, a delayed beam acquisition timeout, an incomplete office-table lift and an exhausted living-room route search. Of 40 starts, 37 achieve joint attachment, 36 complete lift, and 35 obtain and execute a verified route (supplementary stage counts). The conditional 35/35 transport result applies to successfully acquired states; acquisition remains part of team reliability. Four trials per scene support descriptive counts, not precise scene-level reliability estimates.

For 31 successful doorway missions, median planned/measured minimum clearances are 32.5/29.0 mm, with a median within-trial difference of 2.2 mm. One reaches 19.95 mm: above the execution threshold but below the planning margin. These measurements concern control-step payload/base footprints, not continuous clearance of the

articulated robot geometry. Among successful missions, median first goal arrival is 95.4 s and completed lift 10.6 s; transport accounts for a median 88.6% of duration (supplementary duration analysis).

Median search and smoothing times are 5.15 and 19.83 s; their per-mission sum has median 26.45 s (11.41–70.70 s), excluding verification and retries. These wall-clock costs are excluded from simulated completion times. Configured 10 Hz MPC and 50 Hz actor rates therefore do not establish real-time execution.

F. Formation variation and operational timing

Across the 35 successful records, base-offset deviation from first completed lift has median per-trial maximum 16.2 mm and overall maximum 40.8 mm. The latter exceeds the nominal 20 mm margin, so Eq. (3) does not uniformly certify these executions. Measured clearance remains an empirical reduced-model check.

Global search and verification pause simulation after lift; active holding during planning is untested. The logs omit complete MPC solve-time and deadline-miss distributions. Zero twist after a rejected solve is a fallback, not a safe-recovery guarantee.

G. Spatial reorientation and later evaluations

The panel probes use a 1.5 kg, $1.90 \times 0.09 \times 1.25$ m object, supervised acquisition and welded retention. A 0.15 m clearance lift precedes roll; reference durations are 22.5/45 s for 45°/90°. The geometric teacher reaches 44.29° and 88.29°, respectively (Fig. 3). Passing requires 3 s of terminal holding with position error below 25 mm, geodesic rotation error below 3°, linear/angular speed below 0.04 m/s and 0.04 rad/s, retained contacts and no physical termination. Only positive roll is tested. These are selected configuration-specific probes, not a success-rate estimate across the envelope in Eq. (10).

The separate spatial imitation actor completes the 0° and 15° probes of a three-angle, same-start test; the 45° probe terminates on robot self-penetration at 17.26° measured roll. The successful 15° probe ends at 12.83° (supplementary rotation traces). Teacher and learned probes use different starts and handle spans (0.20 versus 0.16 m); they do not isolate a controller effect. A historical 180° teacher attempt fails at 51.40° on the tracking guard. It remains in the supplementary record as an out-of-envelope diagnostic, not proof of a sharp physical threshold at 90°.

Two additional prespecified teacher probes use one rail per carrier (vertical rail separation zero): a crate and a beam, each 1.5 kg, commanded to 15° with a 15 s rotation segment after clearance lift. They complete at 14.92° and 14.15°, respectively, under the same terminal tolerances. These two nominal probes demonstrate small-angle rotation with single-bar grasps; they do not identify a maximum safe angle or isolate the effect of rail geometry across different objects.

A later frozen planar cohort completes 46/48 missions across twelve scenes, using a different actor and upper/lower panel grasps. Both failures are acquisition timeouts. This

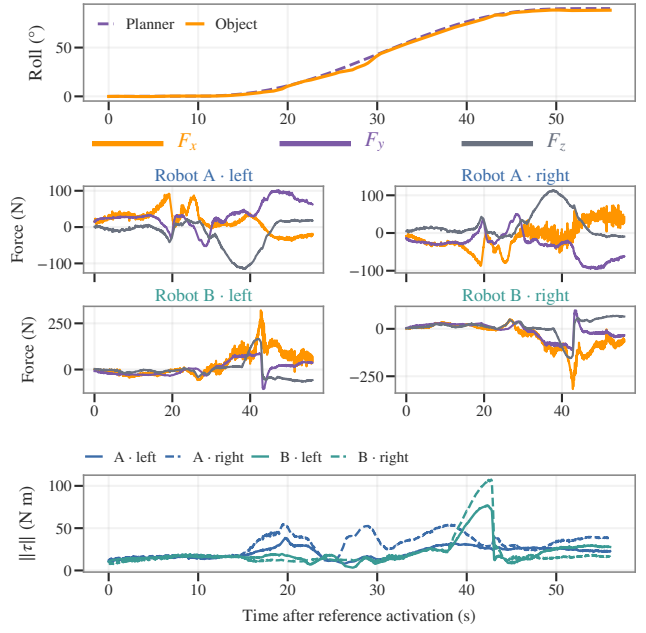


Fig. 4. Planner reference, measured object motion and per-hand reactions. The same pose-to-wrench interface applies to each hand; this diagnostic uses one instrumented 90° panel trial with the geometric teacher. Force components use the fixed lifted-object axes; moment norms are palm-centered. Dashed roll is commanded, while hand loads are measured weld reactions rather than force targets. All panels share the time axis; force scales vary. Full moment components are supplementary.

broader cohort retains supervised progression and welds; its checkpoint and domain differ from the 35/40 cohort, so the scores are not a matched improvement estimate. The supplement reports its full scene counts and the completed local-history recovery comparisons. Neither cohort evaluates large-angle rotation during doorway passage.

H. From object commands to hand reactions

A separate instrumented teacher probe repeats the nominal 90° roll with a 1.5 kg panel (seed 207000). The planner commands pose and twist, not hand wrenches. For hand h of robot i , selected weld multipliers λ_{ih} give the payload wrench through

$$J_O^\top w_{ih}^{O,W} = S_O J_{ih}^\top \lambda_{ih}, \quad w_{ih} = \begin{bmatrix} R_0^\top F_{ih}^W \\ R_0^\top [\tau_{ih}^{O,W} - (p_{ih}^W - p_O^W) \times F_{ih}^W] \end{bmatrix}. \quad (24)$$

Here S_O selects the free payload's six generalized coordinates, J_O maps their velocities to its world twist, and J_{ih} is the weld Jacobian. R_0 fixes the lifted-object axes; moments act about the measured palm center p_{ih} . These solver reactions act on the payload and exclude separate finger-contact forces.

The probe completes at 88.29° (Fig. 4). Over 2,802 control samples, left/right force-norm RMS values are 73.1/71.7 N for Robot A and 94.7/97.1 N for Robot B. Summed hand forces match the separately logged robot totals within 10^{-6} N. Opposing reactions can therefore be obscured by aggregate load feedback: one hand reaches 363.2 N and another 107.4 N m despite the light payload. Successful pose tracking alone does not establish feasible grip loads. These weld-supported

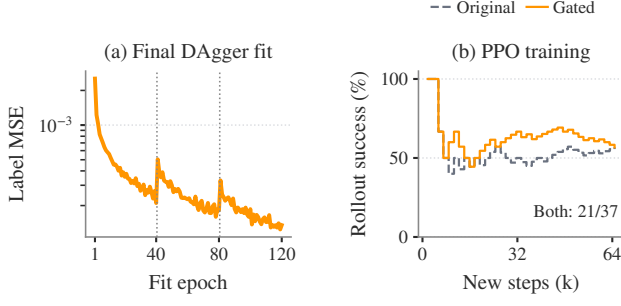


Fig. 5. Archived imitation and separate pickup PPO training traces. The two panels correspond to different learning procedures and evaluations.

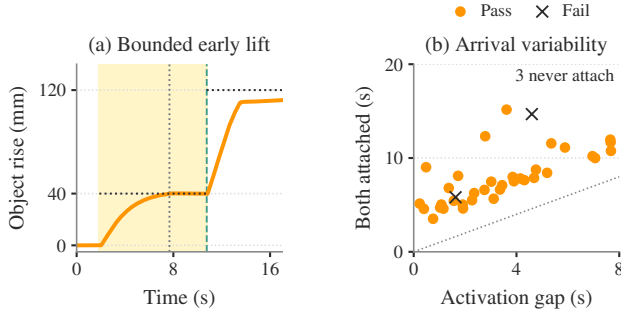


Fig. 6. Independent arrival. (a) Loading dock, seed 142013; measured object rise under a 7.68 s activation gap. Shading marks early lift; vertical dotted/dashed lines mark late activation/second attachment; horizontal lines mark height targets. (b) Both-attachment times for the 37 trials that acquire the object; color marks final mission outcome. The diagonal is $t = \Delta$. The other three trials never attach both robots.

loads motivate frictional retention and actuator-limit tests; this probe does not measure wrench tracking or learned intent inference.

VII. DIAGNOSTIC ANALYSES

The following traces complement aggregate outcomes by separating acquisition, geometric feasibility, reference tracking and physical interaction. They retain each recording’s controller and contact model.

A. Learning traces

Imitation loss and stochastic pickup returns describe different optimization procedures; neither replaces frozen mission evaluation.

B. Asynchronous acquisition

Delayed arrival produces a supervised early-lift interval. The trace measures the supervisor-mediated transition, not locally inferred partner readiness.

C. Assembly feasibility

The attained carrier offsets constrain admissible yaw more strongly than the payload alone.

D. Timing and model consistency

The paired development trace illustrates a planning inconsistency; the simultaneous geometry and margin changes prevent causal attribution to either alone.

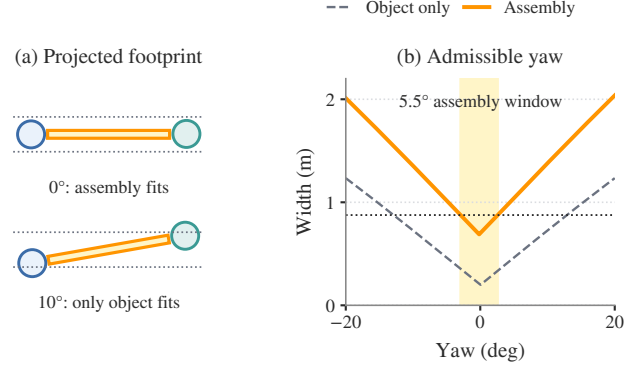


Fig. 7. Why the grasp changes passage feasibility. Analytic projections use the measured beam handoff geometry (seed 142024). (a) Dashed bounds show the available opening width after the 22 mm search margin on each side; the circles are base footprints. (b) The object-only yaw range is about 25.8° , versus 5.5° for the assembly (shaded); the horizontal dotted line is the door budget.

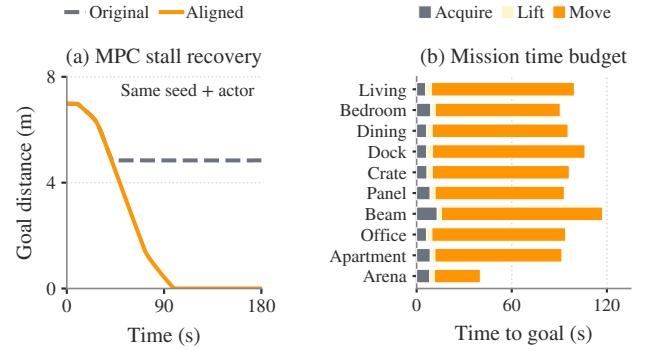


Fig. 8. Planning consistency and mission duration. (a) Same-seed, same-actor MPC comparison (114000): the original model stalls before the doorway; aligned geometry and margin permit progress to the goal. (b) Mean phase durations among successful frozen missions in each scene. Navigation dominates elapsed simulation time. The paired development case and frozen test use different actors and domains.

E. Stage attrition and clearance

Stage counts retain all starts. Reduced-model execution clearance is measured separately from planned clearance.

F. Spatial motion sequences

Panel rotation is a fixed-grasp teacher probe. It does not demonstrate hand switching or learned control over the full angular envelope.

G. Orientation tracking

Requested endpoints, evolving references and measured orientation are distinct quantities; failed and out-of-envelope attempts remain visible.

H. Command-to-interaction response

Pose and twist are commanded at object level. The hand wrenches are resulting solver reactions, not force commands. All four hands are measured in one teacher replay.

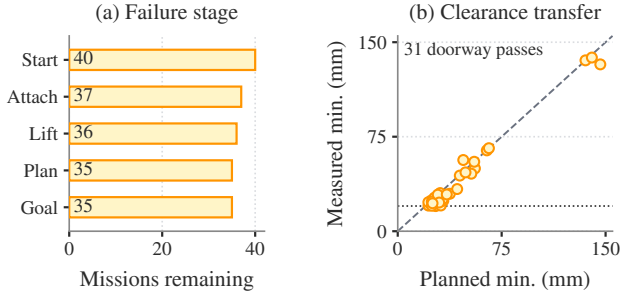


Fig. 9. (a) Mission counts at each stage, retaining all 40 starts. (b) Planned versus measured minimum clearance for 31 successful doorway missions; four arena passes have no finite clearance. Diagonal: equality; horizontal dotted line: 20 mm planning requirement.

I. Retained native-contact failures

Figure 13 pairs failed acquisition states with measured palm errors and contact qualification. These three cases belong to the older 26-trial pre-fix development screen: self-penetration terminates the couch and crate attempts, while unsupported object travel terminates the beam attempt; all also record planning failure. They are diagnostic recordings, not acceptance evidence for the current controller. The latest exported frozen acquisition cohort completes 0/20 trials and terminates on unsupported object travel. Its task is approach, four-hand acquisition, a 6 cm lift and an 8 s hold; it does not evaluate full transport. Small palm error alone therefore cannot be interpreted as successful grasping or load support.

VIII. DISCUSSION

A. Geometry, control and physical scope

The aperture analysis shows that assigned carrier placement restricts object-path feasibility. The formation bound identifies a second requirement: execution must preserve enough of the planned layout to retain clearance. Observed drift prevents a uniform certificate at the nominal 20 mm margin. A geometry-by-margin experiment and ordinary versus aperture-biased RRT-Connect under equal budgets are still needed to isolate planning benefits.

The 35/40 and 46/48 cohorts establish operation of supervised stacks, not the advantage of learned control. A matched geometric-teacher baseline must hold starts, planner, supervisor and contact model fixed. Spatial probes sharpen this distinction: teacher-controlled 90° rotation is demonstrated, whereas the learned 45° probe fails. Upper/lower grasp geometry alone does not establish learned reorientation capability. Rolls beyond 90° require hand reassignment to avoid arm twisting and are excluded.

Welded retention, light loads and simulator-derived feedback limit physical interpretation. Joint-limit and selected collision guards do not certify the full articulated swept volume. The four-hand diagnostic exposes large opposing weld reactions despite successful motion tracking. Frictional retention, actuator saturation and hardware sensing require separate validation. The newer breakable-grasp local-history trials in the supplement expose recovery failures; they neither replace the welded cohorts nor establish broad robustness.

Evaluation within development scene families and one training seed per actor also limits claims of generalization.

B. Shared intent and local feedback

The broader questions remain **RQ1**: how much must be shared beyond local object feedback; **RQ2**: which message content matters at a fixed budget; and **RQ3**: when task-level agreement supports local control under changing partner behavior. Central training can exploit joint state without providing it to executing actors. Conversely, local actors can receive joint readiness implicitly through explicit shared references. These are distinct information boundaries.

The supervised cohorts and preliminary local-history trials use different contact models, observations and tasks. They are not a communication ablation. Testing the questions requires matched message-content and rate sweeps, locally inferred versus supervised progression, and accounting for shared perception traffic as well as direct robot messages.

IX. CONCLUSION

Shared weights leave a team to coordinate where it can move and when it can begin. Grasp-conditioned assembly planning and shared learned local control complete 35/40 frozen missions and 46/48 in a separate later cohort. Carrier-aware apertures and measured formation drift link common geometry to execution. Spatial probes further demonstrate teacher-controlled fixed-grasp roll up to 90°, while retaining the learned controller’s failures. Together, these results define a supervised planning–control baseline; matched experiments remain necessary to establish when task-level intent suffices for reliable local coordination.

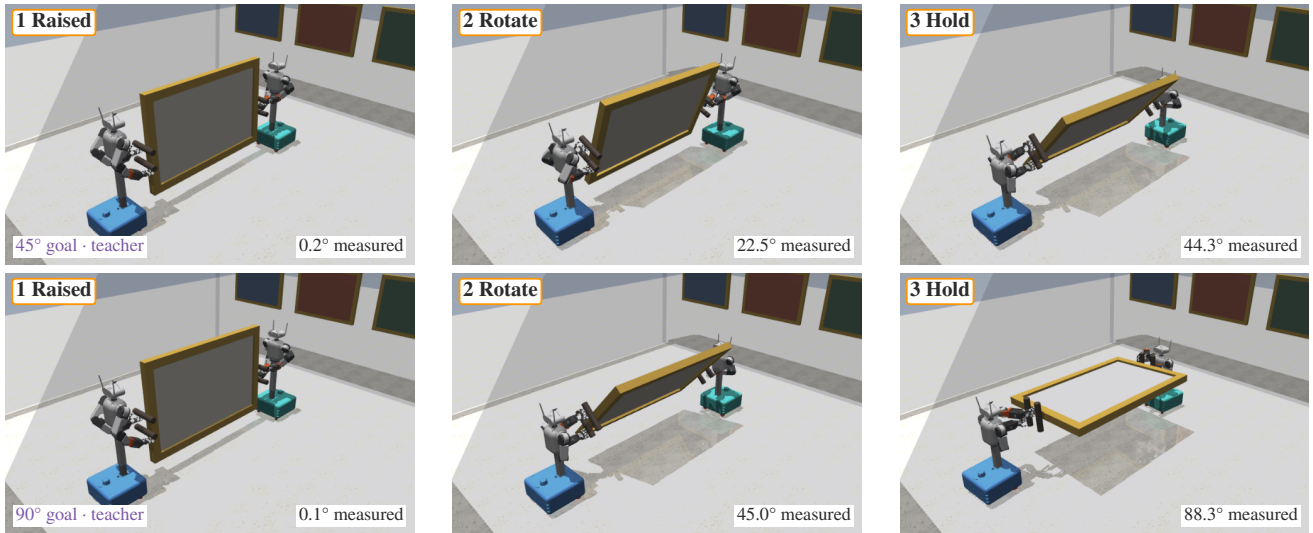


Fig. 10. Recorded fixed-grasp panel reorientation with the geometric teacher (seed 207000). Rows request 45° and 90° roll about the lifted object’s long axis. Columns show clearance lift, intermediate rotation and terminal holding; labels give measured angles. Robot A/B have blue/teal bases and native body materials. Original joint states are rendered without stepping dynamics. These are pose-only probes, not learned-policy or doorway-transport successes.

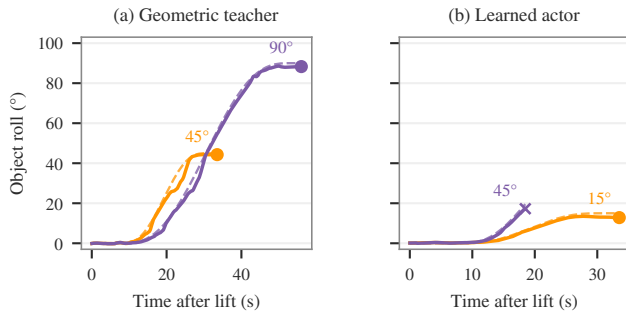


Fig. 11. Measured roll (solid) and evolving reference (dashed), relative to the lifted object frame. Circles mark successful completion; the cross marks self-penetration termination. Labels specify commanded endpoints, which are distinct from the evolving reference. Teacher and actor configurations differ.

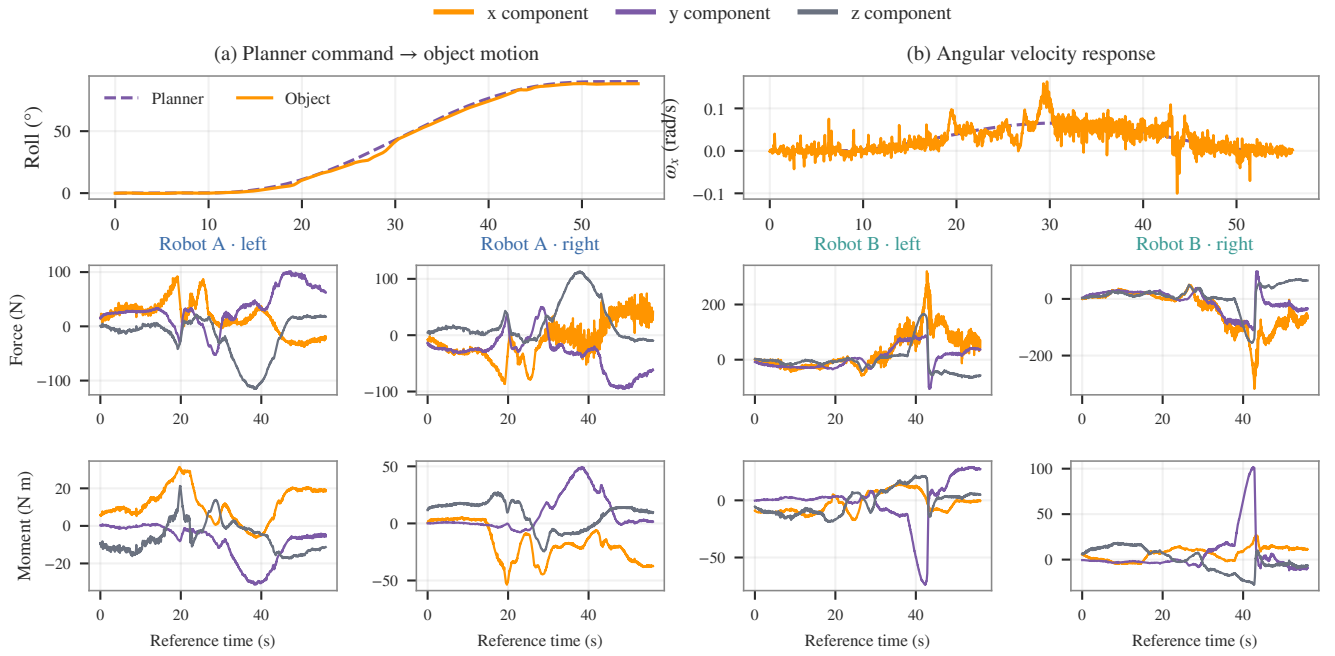


Fig. 12. Commanded object motion and resulting four-hand weld reactions in one instrumented 90° teacher probe. Dashed purple: pose/twist reference; orange: measured object response. Hand components use fixed lifted-object axes (x : orange, y : purple, z : gray); moments are palm-centered. Time starts at pose-schedule activation; ordinate scales vary by panel. These are simulated reactions, not commanded force targets or wrist-sensor data.

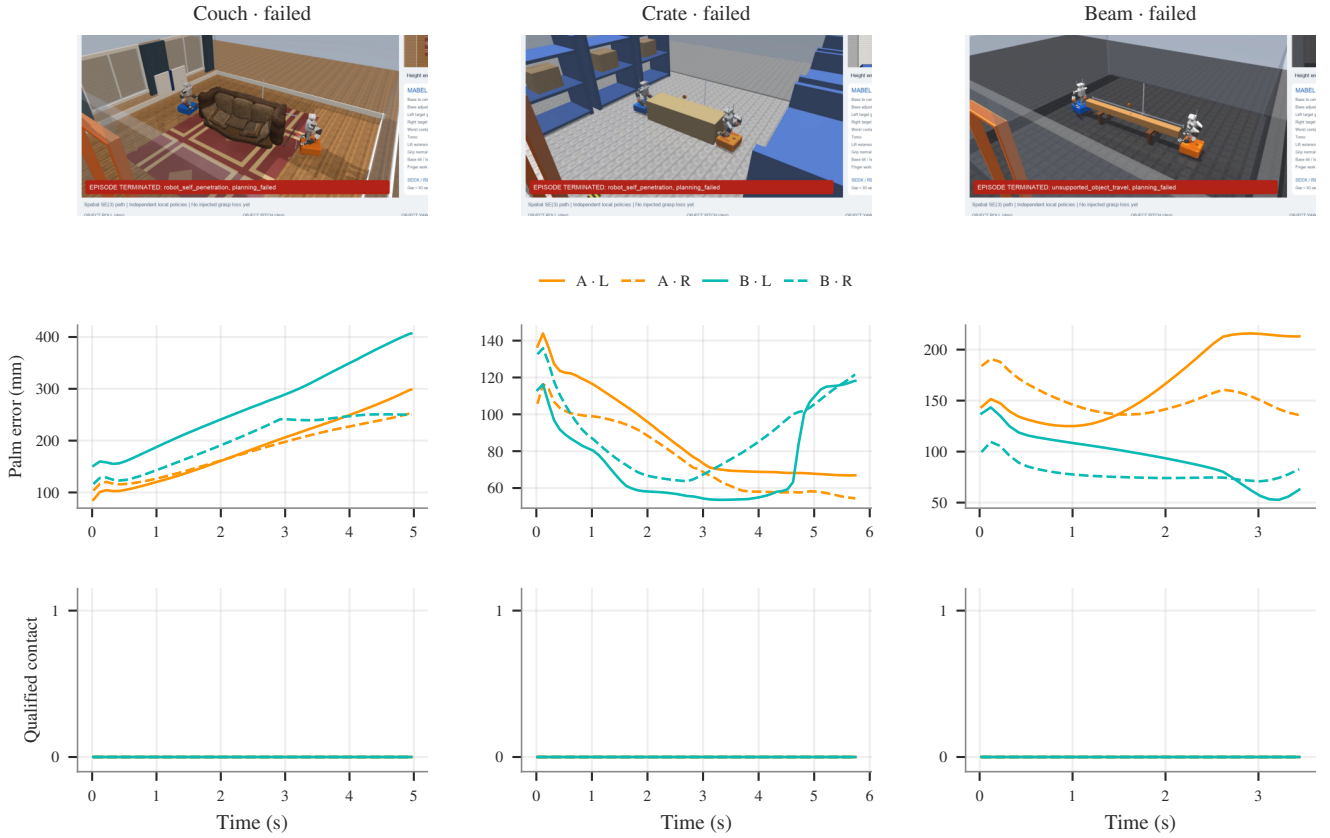


Fig. 13. Retained native-contact acquisition failures: recorded terminal states, palm-target distance and raw contact qualification. Seeds 500000/500002/500006; pre-fix development screen.

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